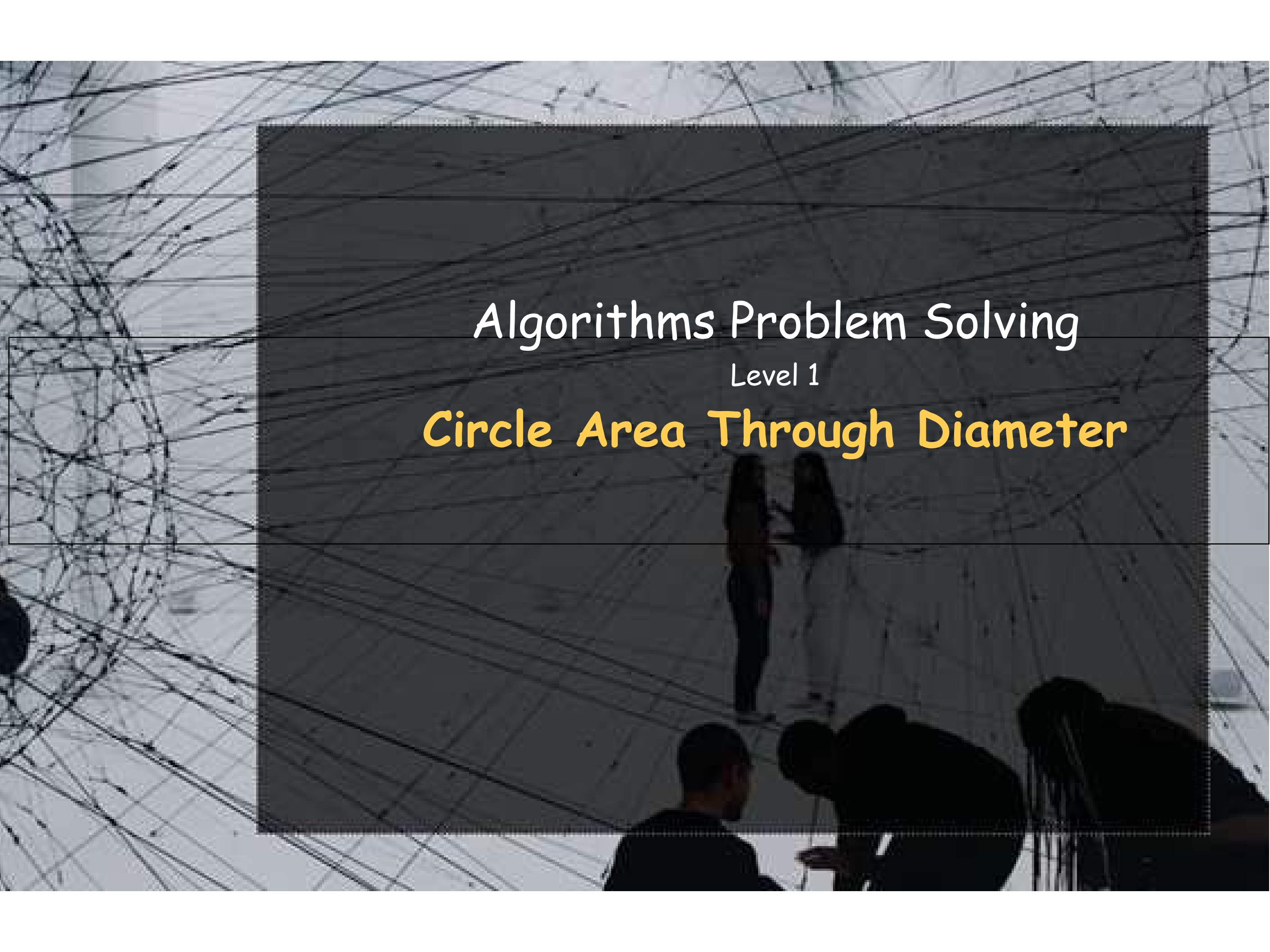


Algorithms Problem Solving

Level 1

Circle Area Through Diameter



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Definition

- ◆ This program reads the diameter of a circle and calculates its area.
- ◆ The program uses functions, the mathematical constant π , `pow()`, division, and float values.
- ◆ Since the diameter is given instead of the radius, the formula is based on the relationship between diameter and radius.

Explanation

- ◆ The `readNumbers()` function reads the circle diameter `D` from the user.
- ◆ The radius is half of the diameter:
- ◆ $\text{Radius} = D / 2$
- ◆ The normal circle area formula is:
- ◆ $\text{Area} = \pi \times R^2$
- ◆ Substituting $R = D / 2$ gives:
- ◆ $\text{Area} = \pi \times (D / 2)^2$
- ◆ This can be simplified to:
- ◆ $\text{Area} = \pi \times D^2 / 4$
- ◆ The `CircleArea()` function uses this formula:
- ◆ `M_PI` represents π .
- ◆ `pow(D, 2)` calculates D^2 .
- ◆ The result is divided by 4 to account for the diameter being twice the radius.
- ◆ The `printArea()` function receives the calculated area and prints it.
- ◆ In `main()`, the return value of `CircleArea(D)` is directly passed to `printArea()`.

Steps in main()

1. Read the circle diameter D .
2. Calculate D^2 using `pow()`.
3. Multiply D^2 by π .
4. Divide the result by 4.
5. Return the calculated area.
6. Pass the returned area to `printArea()`.
7. Print the circle area.
7. Print the appropriate result.

💡 Example

◆ If the user enters:

Diameter: 10

◆ First, calculate the radius:

$$\text{Radius} = 10 / 2 = 5$$

◆ Then calculate the area:

$$\text{Area} = \pi \times 10^2 / 4$$

$$\text{Area} = \pi \times 100 / 4$$

$$\text{Area} = \pi \times 25$$

◆ The result is approximately 78.5398.

➡ Result

Circle Area = 78.5398

Note

- ◆ The formula $\pi \times D^2 / 4$ is derived from the standard formula $\pi \times R^2$.
- ◆ D represents the diameter, while R represents the radius.
- ◆ The code uses int for the diameter and float for the calculated area.

⊘ Warning

- ◆ The program assumes the diameter is a valid positive value. It does not check whether the user enters zero or a negative number.
- ◆ "Ciecle" is misspelled in the output text. It should be "Circle".



Comparison

▨ Using Radius

- ✓ Formula: $\text{Area} = \pi \times R^2$.
- ✓ The radius is given directly.
- ✓ No division by 4 is needed.

▤ Using Diameter

- ✓ Formula: $\text{Area} = \pi \times D^2 / 4$.
- ✓ The diameter is given directly.
- ✓ Division by 4 is needed because $D = 2R$.

✦ Summary

- ✓ A circle's area can be calculated directly from its diameter.
- ✓ The formula using diameter is $\pi \times D^2 / 4$.
- ✓ The diameter is twice the radius.
- ✓ `pow(D, 2)` calculates the diameter squared.
- ✓ A function's return value can be passed directly to another function.

Thank you 😊